

Final Report (PDF)

Internet Adoption and Economic Growth: A Comparative Analysis of Global Development (2000–2024)

AnalystLab Africa – Week 8 Capstone Project

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1. Introduction

Background

The internet has become a key driver of economic development, enabling innovation, improving productivity, expanding access to markets, and accelerating digital transformation. Governments and international organizations continue to invest in digital infrastructure to promote inclusive economic growth.

This project investigates the relationship between internet adoption and economic performance using data from the World Bank's World Development Indicators (WDI). By analyzing trends across countries and regions between 2000 and 2024, the project aims to determine whether increased internet adoption is associated with higher levels of economic prosperity.

2. Project Objective

The primary objective of this project is to analyze the relationship between internet adoption and GDP per capita across countries using World Development Indicators.

Specifically, the project aims to:

- Analyze global internet adoption trends between 2000 and 2024.
 - Compare internet adoption across regions and income groups.
 - Examine GDP per capita trends over the same period.
 - Investigate the relationship between internet adoption and GDP per capita.
 - Present findings using an interactive Power BI dashboard.
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3. Dataset Description

Dataset Source

World Bank – World Development Indicators (WDI)

The WDI dataset contains development indicators for over 200 countries and territories covering themes such as economic performance, education, health, technology, trade, and infrastructure.

Indicators Selected

- Individuals using the Internet (% of population)
- GDP per capita (current US\$)
- Population, total

Coverage

- Countries: 217
 - Regions: 7
 - Years: 2000–2024
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4. Data Cleaning Process

Python (Pandas) was used to clean and prepare the dataset before visualization.

The following steps were completed:

Indicator Filtering

Only indicators relevant to the project objective were retained:

- Internet Adoption
- GDP per Capita
- Population

This significantly reduced the size of the original dataset.

Year Filtering

The analysis was limited to the period between **2000 and 2024**, as internet adoption data before 2000 was sparse and less relevant to modern digital development.

Data Transformation

The original dataset stored each year as a separate column. Using the **melt()** function in Pandas, the dataset was transformed into a long format with:

- Country
- Indicator
- Year
- Value

Metadata Integration

Country metadata from **WDICountry.csv** was merged into the cleaned dataset to include:

- Region
- Income Group

Removal of Aggregate Records

Regional aggregates and non-country entities were removed to ensure that only sovereign countries were included in the analysis.

Missing Values

Missing values in the **Value** column were intentionally retained because they represent unavailable or unreported observations rather than zero values.

Income Group values for a small number of countries remained blank because they were unavailable in the World Bank metadata.

Data Type Validation

The Year column was converted to an integer, while Value was maintained as a decimal numeric field suitable for analysis in Power BI.

5. Analysis Methodology

The project followed the complete data analytics workflow.

Tools Used

- Python (Pandas)
- Jupyter Notebook
- Power BI

Analytical Techniques

- Data Cleaning
 - Data Transformation
 - Trend Analysis
 - Comparative Analysis
 - Correlation Analysis
 - Interactive Dashboard Development
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6. Dashboard Overview

The Power BI dashboard was designed to communicate the relationship between internet adoption and economic performance.

The dashboard includes:

- KPI Cards
 - Internet Adoption Trend
 - GDP per Capita Trend
 - Scatter Plot showing Internet Adoption vs GDP per Capita
 - Top Countries by Internet Adoption
 - Regional Analysis
 - Income Group Analysis
 - Interactive Slicers
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7. Key Findings

The analysis produced several important findings:

Internet adoption increased significantly

Average internet adoption increased steadily between 2000 and 2024, demonstrating rapid global digital transformation.

High-income countries lead digital adoption

Countries classified as high income consistently recorded the highest internet adoption rates.

Positive relationship between internet adoption and GDP per capita

The scatter plot revealed a positive relationship between internet adoption and GDP per capita. Countries with higher internet penetration generally exhibited higher levels of economic prosperity.

Regional disparities remain

North America and Europe & Central Asia consistently outperformed other regions in both internet adoption and GDP per capita.

Emerging regions show strong progress

Sub-Saharan Africa experienced rapid internet adoption during the study period, although GDP per capita remained comparatively lower than in more developed regions.

8. Insights

The findings suggest that digital connectivity plays an important role in economic development.

Countries with widespread internet access benefit from improved communication, increased business opportunities, greater innovation, and stronger participation in the digital economy.

Although internet adoption alone does not determine economic performance, the analysis indicates a clear positive association between digital inclusion and GDP per capita.

9. Recommendations

Based on the analysis, the following recommendations are proposed:

- Increase investment in broadband infrastructure, particularly in developing economies.
 - Improve internet affordability to encourage greater digital inclusion.
 - Promote digital literacy and workforce development.
 - Encourage public-private partnerships to expand digital infrastructure.
 - Continue monitoring internet adoption alongside key economic indicators to support evidence-based policy making.
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10. Conclusion

This project demonstrates the value of data analytics in understanding global development trends. Using World Development Indicators, Python, and Power BI, the analysis identified a positive relationship between internet adoption and GDP per capita across countries between 2000 and 2024.

The interactive dashboard enables policymakers and stakeholders to explore regional and income-based differences, providing valuable insights that can support future digital and economic development initiatives.